

Setting Up Warehouses for Delivery of Goods

T01-D

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Overview of Problem

- **Explanation:** Strategically locating warehouses to better deliver goods
 - **Causes:** Shopping necessities
Requiring efficient fulfillment centers
 - **Expect:** Shortened delivery time and lower costs
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Computational Problem

- ⊙ Computational Objective:

Work out (the number) and locations of warehouses to minimize the total cost

- ⊙ Inputs

- Retailer: coordinate; quantity of demand
- Warehouse: possible coordinate; cost of the construction; how many to be built (p)
- Transportation cost per unit distance

- ⊙ Outputs

- Coordinates of warehouses
 - Total cost
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Computational Problem

⦿ Constraints

- Every retailer can only be served **d** by *one warehouse*
- Every warehouse must be able to meet *the demands of all the retailers* it serves
- Every retailer and warehouse is *beside the road*
- All retailer located within *a specific area*

⦿ Optimization Criteria

- Primary OC: only consider *the distance* (given the number of warehouses)
 - Secondary OC: add *demand* into account
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Algorithm

1. Pre-processing

1. Obtain the coordinates and demand of the retailers.
2. Select the top p retailers with the highest demand as the initial warehouse locations.
3. Assign each retailer to the nearest initial warehouse location to form p clusters.
4. For each cluster, calculate the initial total cost TC_1 .

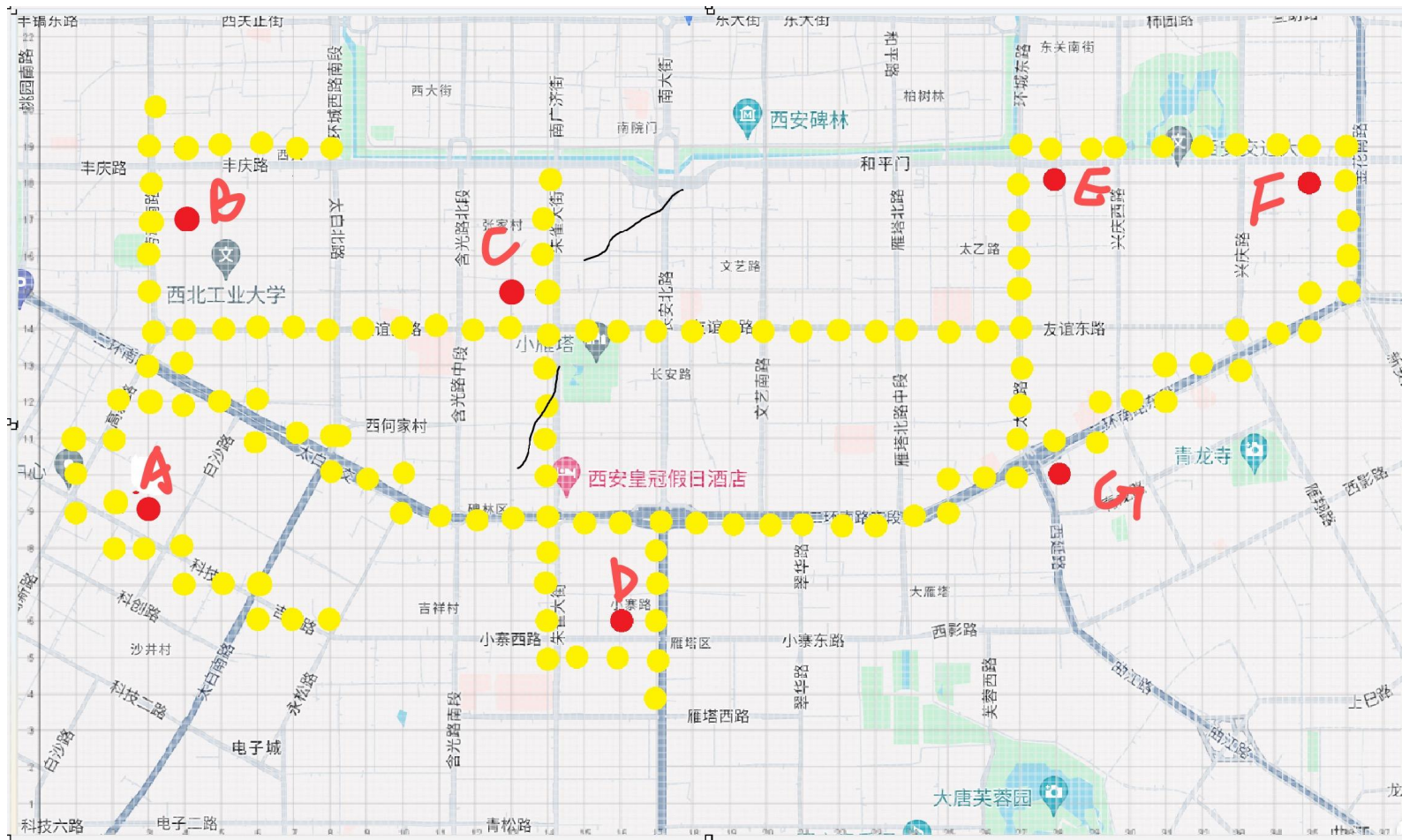
2. Iterative Optimization (performed within each node cluster)

- ◆ Repeat the following steps until the total costs TC of the four neighboring nodes along the road (East, South, West, North) are all greater than the current center location:
 1. Select the **three** neighboring nodes along the road **or on the road** of the current center location.
 2. Calculate the total cost TC for each neighboring node.
 3. Select the node with the smallest total cost as the current solution. **If the current node is on the road, select a node beside the road instead.**
 4. **Calculate and** assign the current total cost to TC_1 .

3. Post-processing

- ◆ Mark the coordinates of the final warehouse locations.

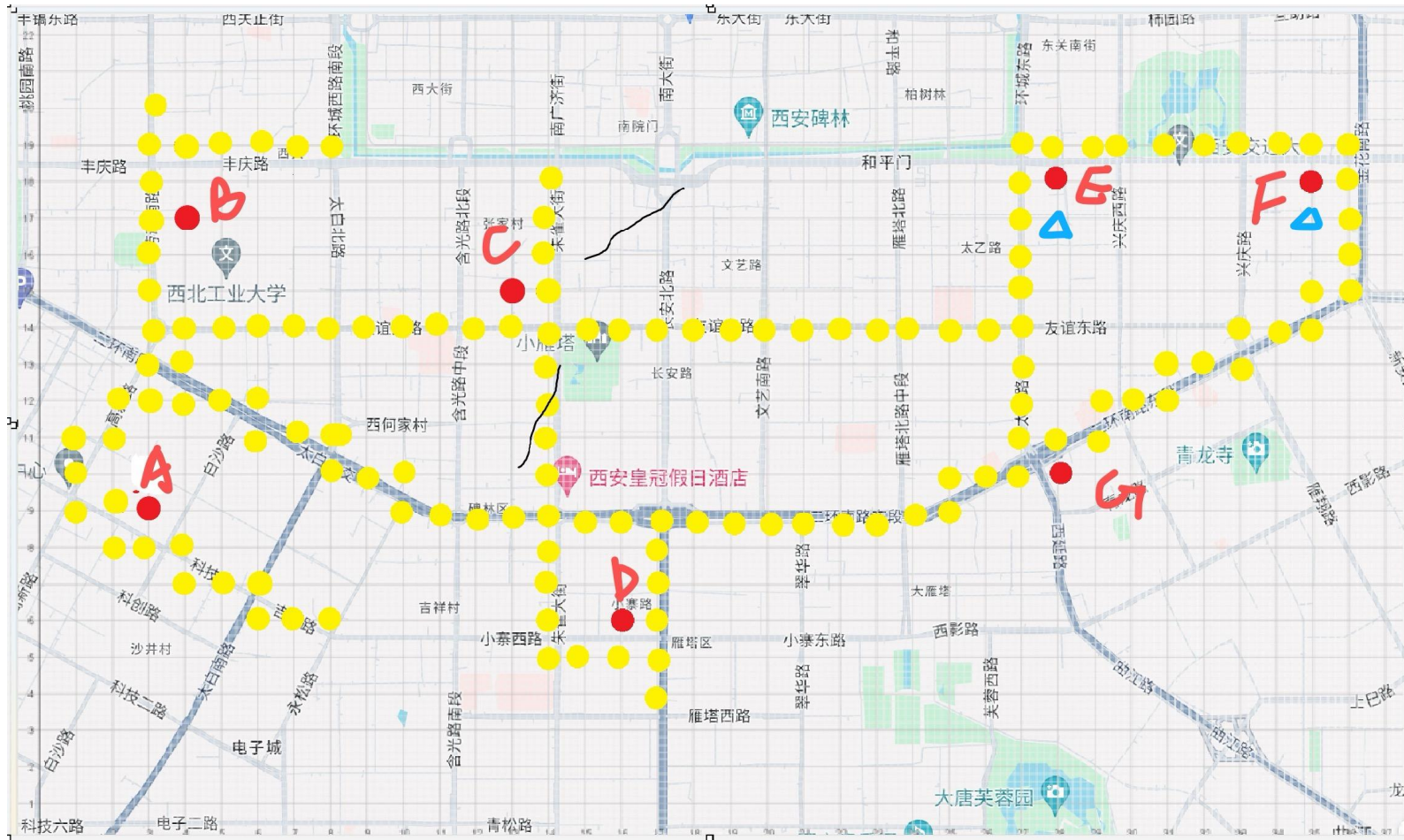
Instance 1



Pre-processing

	Coordinate	Demand
A	(3,9)	250
B	(4,17)	200
C	(13,15)	150
D	(16,6)	100
E	(28,18)	750
F	(35,18)	800
G	(28,10)	700

Instance 1



Pre-processing

Assumption:

- Transportation cost per unit distance is \$1
- The construction cost of each warehouse is the same: \$5000
- The transportation cost per small grid edge length within the warehouse operating time (assuming 5 years) is a constant

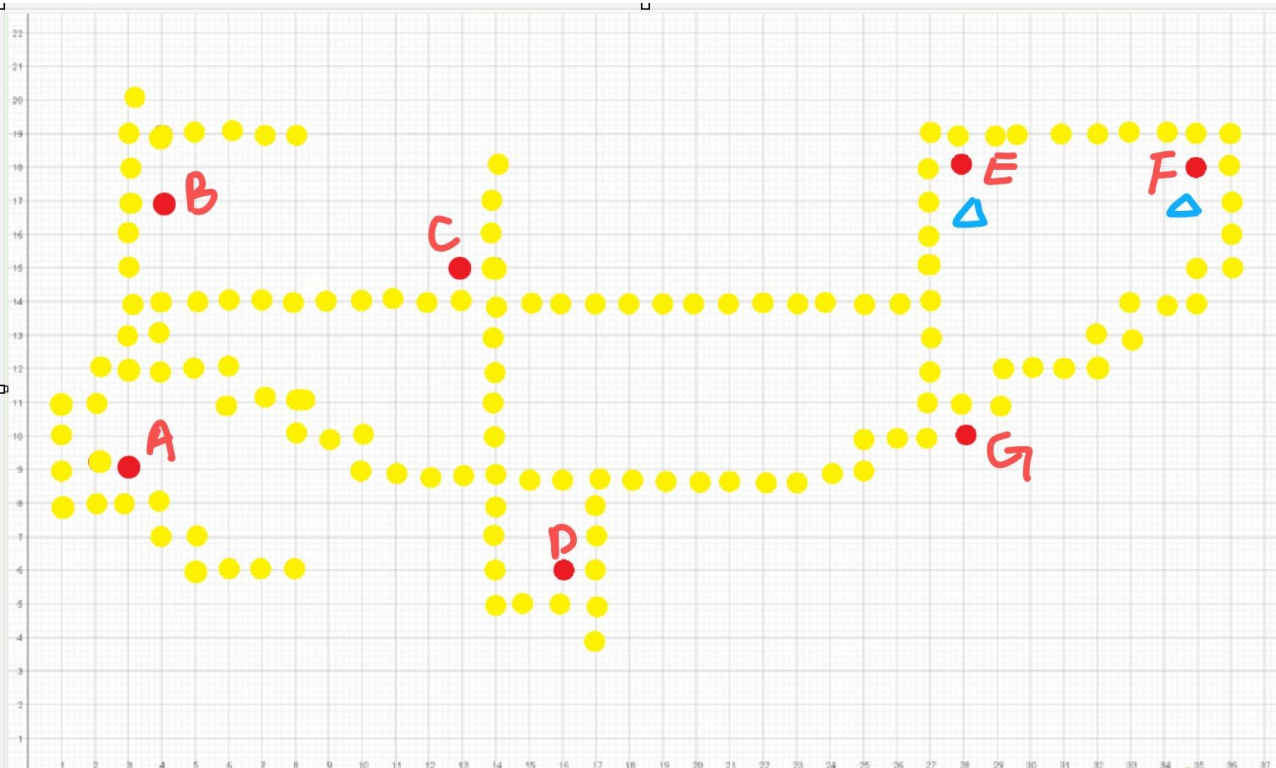
	Coverage warehouse	Total Cost in each cluster
E	For A、B、C、D、 E、G	5000+117
F	For F	5000

Step-by-Step Running of Algorithm

Iterative Optimization (performed within each node cluster):

Repeat the following steps until the total costs TC of the four neighboring nodes (East, South, West, North) are all greater than the current center location:

1. Select the **three** neighboring nodes along the road **or on the road** of the current center location.
2. Calculate the total cost TC for each neighboring node.
3. Select the node with the smallest total cost as the current solution. **If the current node is on the road, select a node beside the road instead.**
4. Assign the current total cost to TC1.

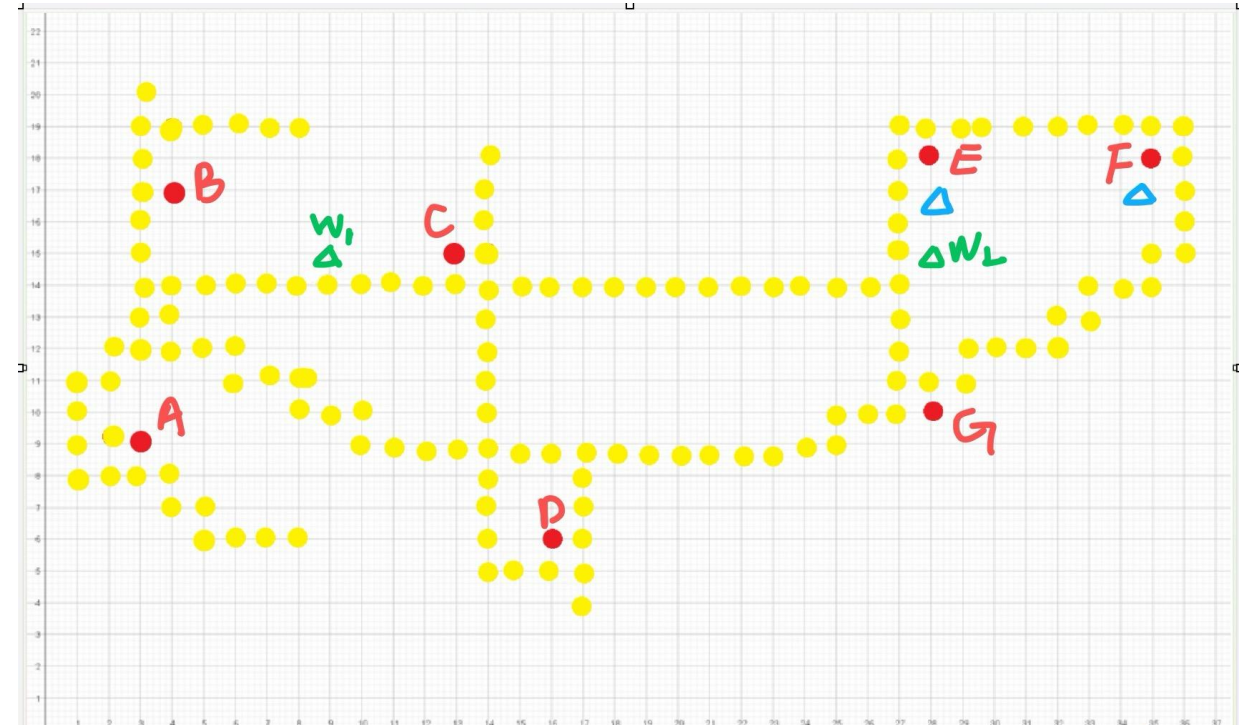


Instance 1

Post-processing:

Mark the coordinates of the final warehouse locations.

We should put an output of our algorithm here. (Maybe not optimal)



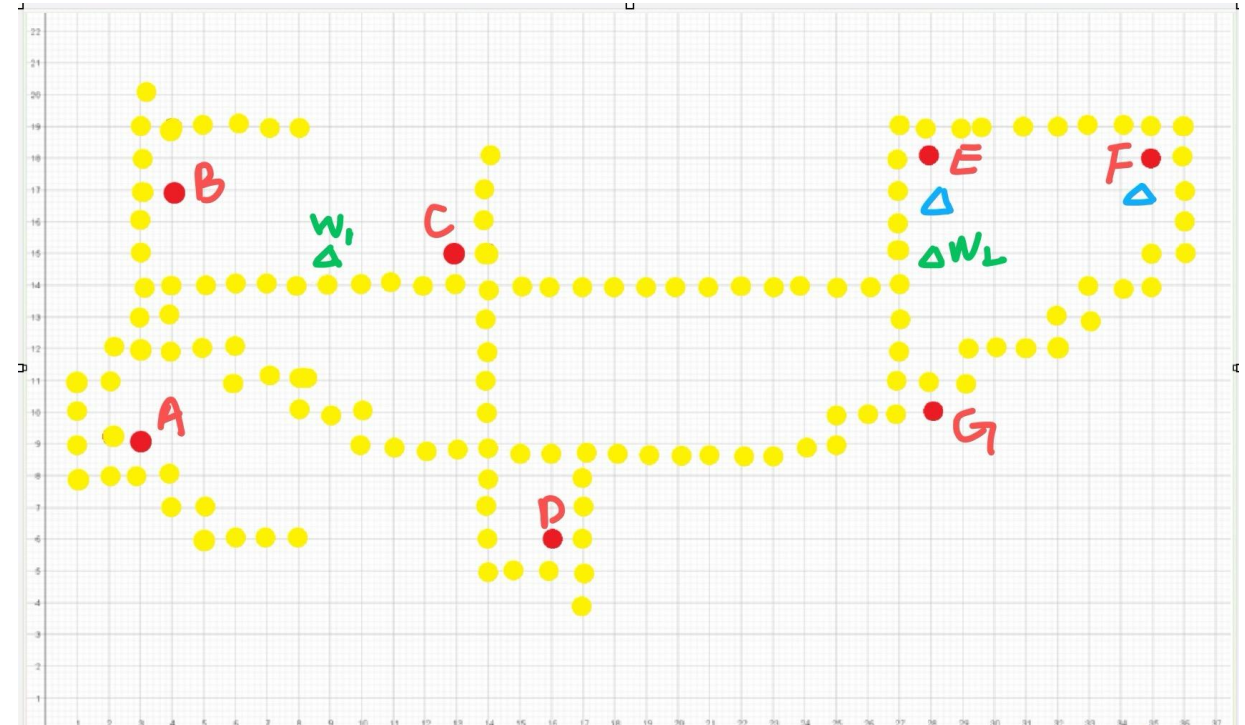
	Coordinate	Coverage warehouse	Total Cost in each cluster
W ₁	(9,15)	For A、B、C、D	5000+22
W ₂	(28,15)	For E、F、G	5000+47

Instance 1

Post-processing:

Mark the coordinates of the final warehouse locations.

This is a more optimal solution, which reveals some disadvantages of our algorithm.



	Coordinate	Coverage warehouse	Total Cost in each cluster
W ₁	(9,15)	For A, B, C, D	5000+22
W ₂	(28,15)	For E, F, G	5000+47

Actual Results

⊙ Coordinates of warehouses

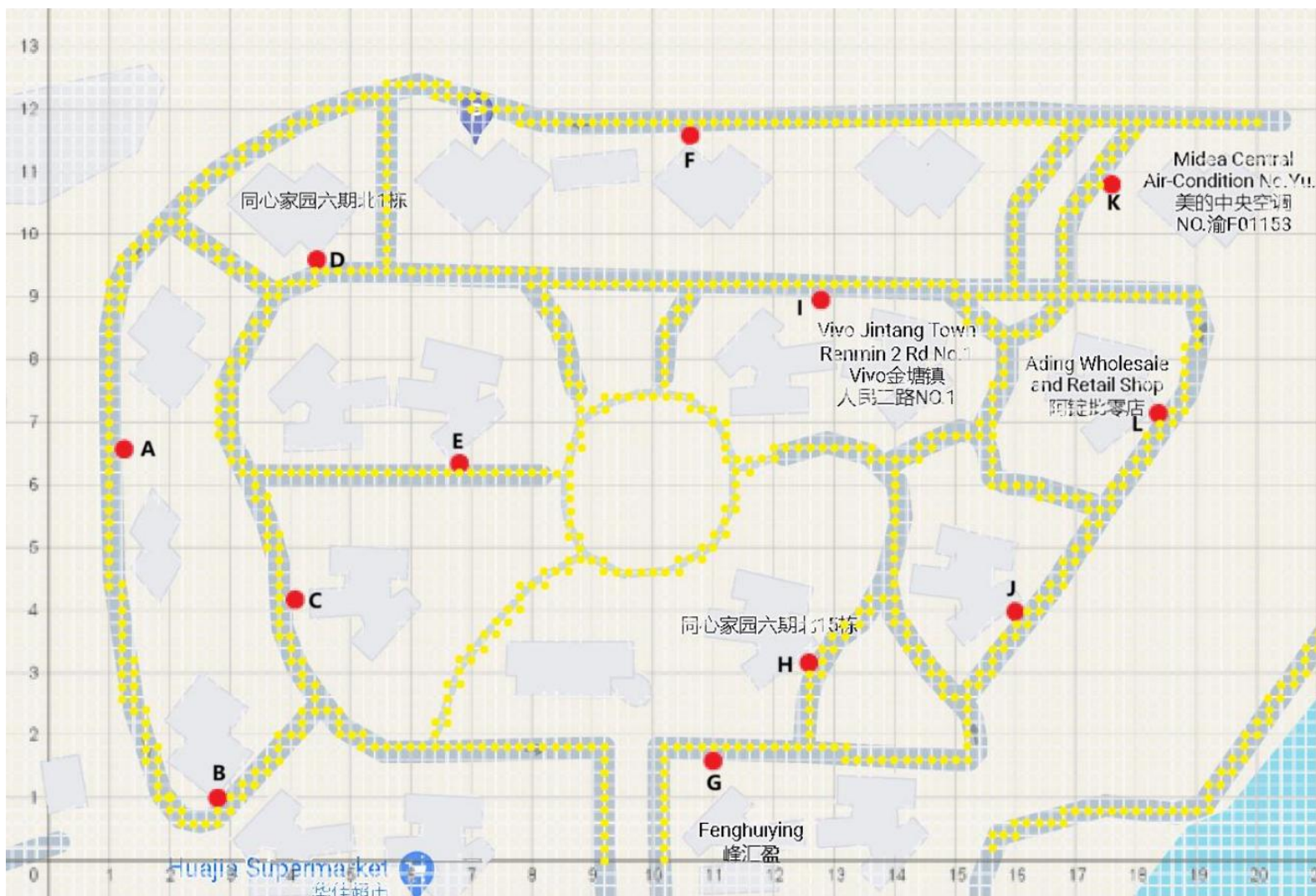
➤ $W_1(9, 15)$ $W_2(28, 15)$

⊙ Total cost

➤ $TC_1 = 10117$

➤ $TC_2 = 10069 < 10117$

Instance 2



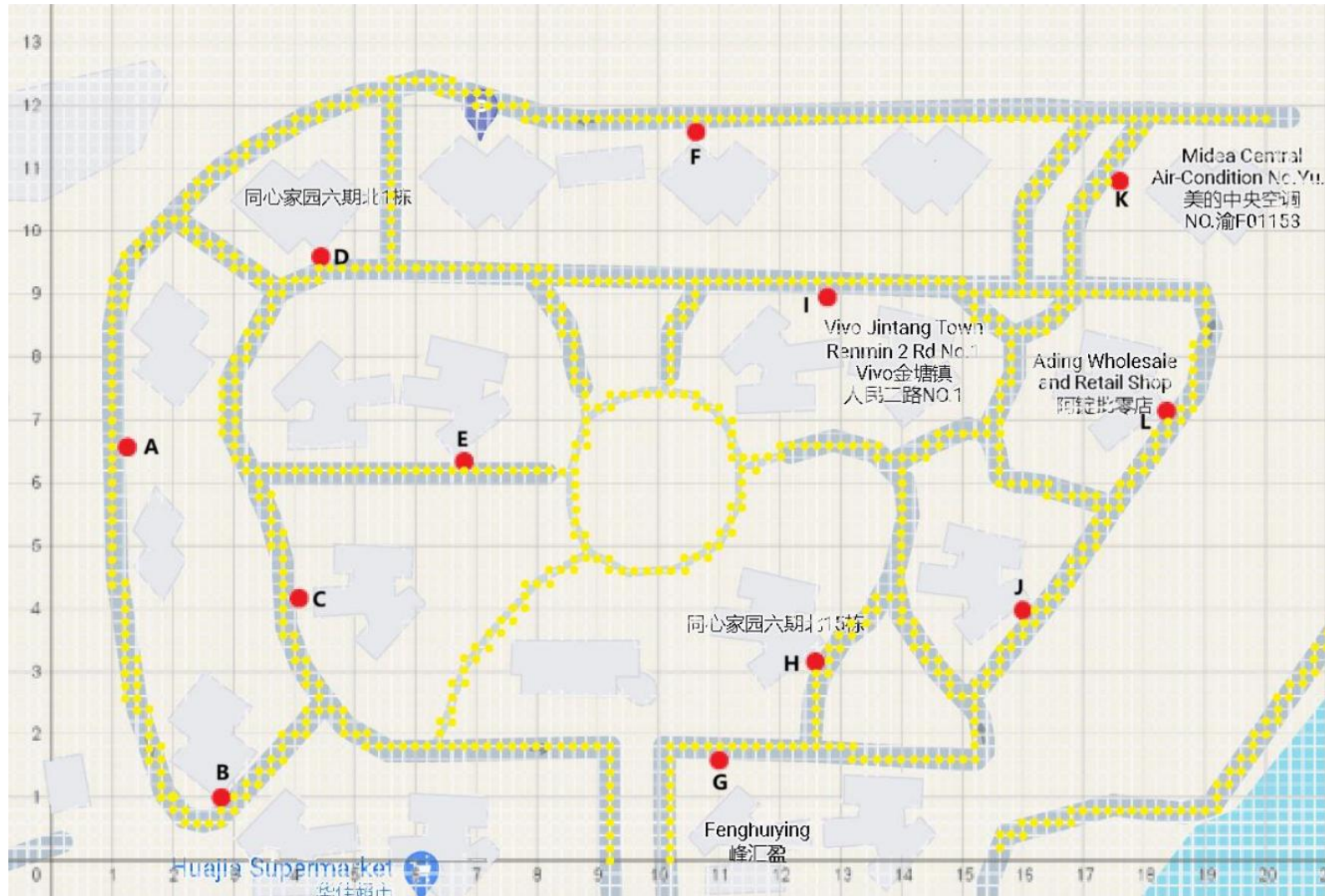
Pre-processing

Assumption:

- Transportation cost per unit distance is \$1
- The construction cost of each warehouse is the same: \$5000

	Coordinate	Demand		Coordinate	Demand
A	(1.2, 6.6)	300	G	(11.0, 1.6)	600
B	(2.8, 1.0)	200	H	(12.6, 3.2)	500
C	(4.0, 4.2)	600	I	(12.8, 9.0)	500
D	(4.4, 9.6)	300	J	(16.0, 4.0)	300
E	(6.8, 6.4)	200	K	(17.6, 10.8)	200
F	(10.6, 11.6)	400	L	(18.4, 7.2)	100

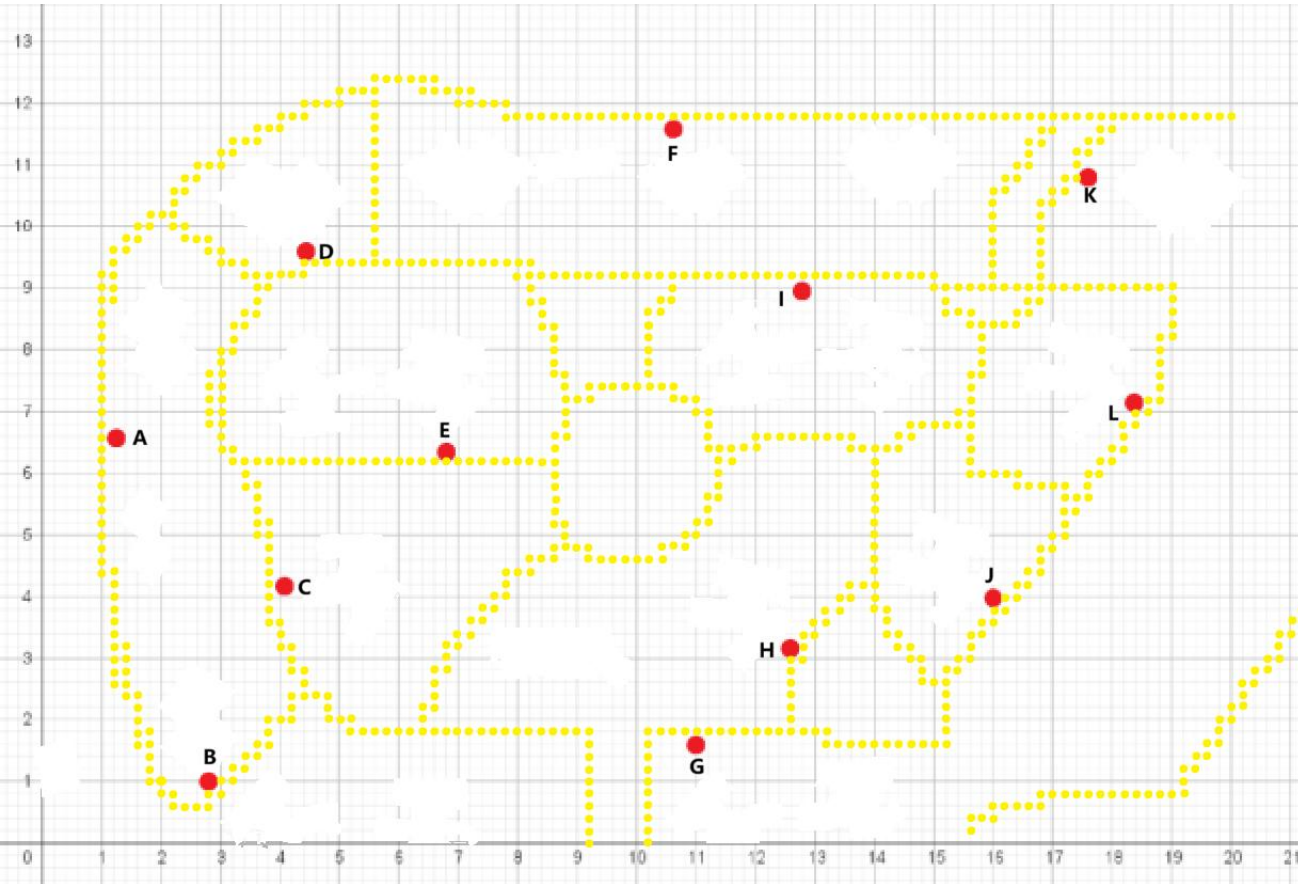
Instance 2



Pre-processing

Initial warehouse	Total Cost in each cluster
C(A, B, D, E)	5000+168
I(F, K, L)	5000+150
H(G, J)	5000+53

Step-by-Step Running of Algorithm

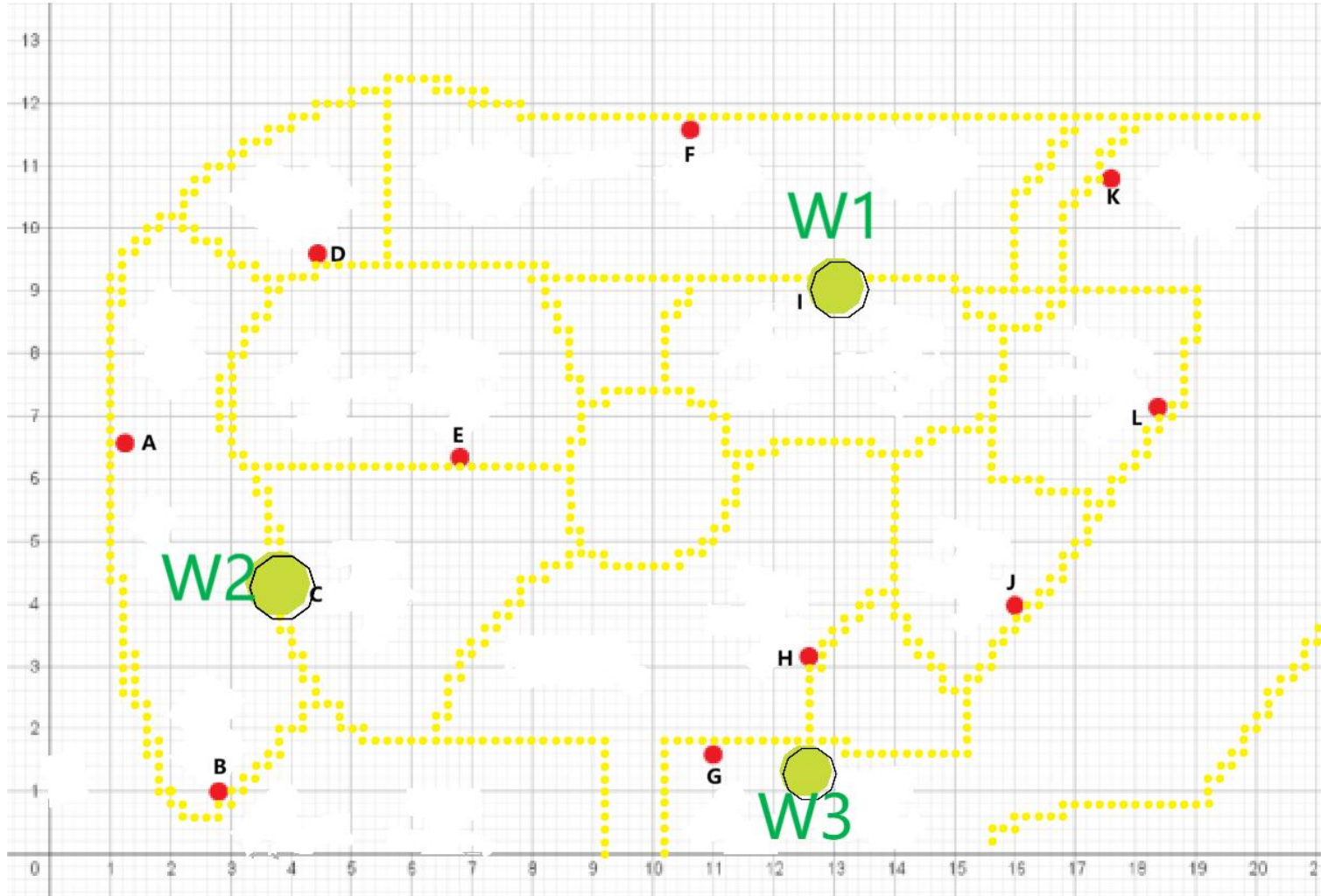


Iterative Optimization (performed within each node cluster):

Repeat the following steps until the total costs TC of the four neighboring nodes (East, South, West, North) are all greater than the current center location:

1. Select the four neighboring nodes along the road (East, South, West, North) of the current center location.
2. Calculate the total cost TC for each neighboring node.
3. Select the node with the smallest total cost as the current solution.
4. Assign the current total cost to TC1.

Instance 2



Post-processing:

Mark the coordinates of the final warehouse locations.

W1(I,F,K): (12.8, 9.0)

W2(A,B,C,D,E): (4.0, 4.2)

W3(G,H,J): (12.6, 1.6)

Actual Results

⊙ Coordinates of warehouses

- $W_1: (12.8, 9.0)$
- $W_2: (4.0, 4.2)$
- $W_3: (12.6, 1.6)$

⊙ Total cost

- $W_1: 5000+40=5040$
 - $W_2: 5000+168=5168$
 - $W_3: 5000+150=5150$
 - $TC = 15358$
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End of Project Report



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